

# Weekly Maintenance & Inspection Report

*Coastal Ridge Wind Farm - Week 37, 2026 (Sep 8-14)*

## 1. Site Overview

Coastal Ridge Wind Farm operated 34 of 36 installed turbines this week, with fleet availability at 96.2%, down from 98.1% last week. Average wind speed was 8.4 m/s across the site. Weather conditions were within normal operating parameters throughout the reporting period, with one high-wind curtailment event on September 11 lasting 90 minutes. Total energy delivered for the week was 4,180 MWh against a forecast of 4,350 MWh, a shortfall attributable primarily to Turbine T-19 being taken offline (see Section 4).

This report covers scheduled inspections, unscheduled findings, consumables and lubrication, health and safety observations, and the maintenance calendar for the coming two weeks. It is intended for the Site Operations Manager, the Mechanical and Electrical Maintenance Leads, and the Health & Safety Officer.

## 2. Turbine Inspection Summary

Routine inspections were completed on all 36 turbines per the standard bi-weekly rotation. The table below summarises turbines with notable readings; turbines not listed returned nominal readings across all checks.

| Turbine | Status      | Vibration (mm/s RMS) | Gearbox Oil Temp (°C) | Notes                                       |
|---------|-------------|----------------------|-----------------------|---------------------------------------------|
| T-03    | Operational | 2.1                  | 61                    | Minor transformer oil weep, monitoring      |
| T-07    | Operational | 2.4                  | 58                    | Brake pad wear at 70%, replace next cycle   |
| T-11    | Operational | 2.0                  | 59                    | Yaw motor bearing noise, within tolerance   |
| T-15    | Operational | 1.9                  | 57                    | SCADA comms dropout, resolved same day      |
| T-19    | OFFLINE     | 8.4                  | 79                    | Gearbox vibration alarm - see Section 4     |
| T-22    | Operational | 2.2                  | 60                    | Harness anchor bolt found loose, see Sec. 6 |

## 3. Mechanical Findings

Turbine T-07's main rotor brake pads were measured at 70% wear during the scheduled inspection.

This is within the acceptable range but approaching the 80% replacement threshold; the mechanical team has scheduled a pad replacement during the next planned maintenance window on September 22.

Turbine T-11's yaw motor exhibited a faint bearing noise during manual yaw testing. Vibration readings remained within tolerance (2.0 mm/s RMS against a 4.0 mm/s RMS alert threshold) and no further action is required at this time beyond continued monitoring at the next inspection cycle.

#### **4. Critical Issue: Turbine T-19 Gearbox Vibration Alarm**

During the Tuesday inspection rotation, Turbine T-19's condition monitoring system recorded a gearbox vibration reading of 7.1 mm/s RMS, exceeding the 6.0 mm/s RMS alert threshold. A follow-up reading on Wednesday measured 7.8 mm/s RMS, and a third reading on Thursday morning measured 8.4 mm/s RMS - three consecutive readings above threshold, each higher than the last. Gearbox oil temperature was also elevated at 79°C against a normal operating range of 55-65°C.

Per the escalation procedure for sustained vibration alarms, the on-site mechanical team initiated an emergency shutdown of T-19 at 11:40 on Thursday, September 11, and notified the gearbox OEM's technical support line the same day. A borescope inspection performed Friday morning identified visible pitting on two intermediate-stage gear teeth, consistent with early-stage surface fatigue rather than a sudden failure.

RECOMMENDATION (CRITICAL, ACTION REQUIRED WITHIN 48 HOURS): The Senior Mechanical Engineer recommends that Turbine T-19 remain offline and that a full gearbox replacement be scheduled rather than an in-field repair, given the OEM's assessment that continued operation risks a catastrophic gearbox failure. The OEM has quoted a replacement lead time of 14 weeks for the standard shipping option. A decision on expedited procurement is needed from Operations Management by September 16, 2026 to avoid further loss of generation revenue. This item is escalated to the Q3 Operations Review meeting (see the accompanying meeting minutes).

#### **5. Electrical Findings**

Turbine T-03's step-up transformer showed a minor oil weep at the base gasket, estimated at less than 50ml over the inspection period. This is below the threshold requiring immediate drain-down and reseal, but has been added to the two-week monitoring watchlist. Turbine T-15 experienced an intermittent SCADA communications dropout on September 9, lasting approximately 40 minutes; the fault was traced to a loose fibre-optic connector at the turbine base cabinet and resolved on-site the same day with no lasting impact on data logging.

#### **6. Health & Safety Observations**

During the pre-climb inspection at Turbine T-22, a technician identified that one of the two internal fall-arrest anchor bolts at the nacelle access hatch was loose - finger-tight rather than torqued to specification. The climb was aborted and the issue was reported immediately to the Site Safety Officer as a near-miss, since the technician had already clipped into that anchor point briefly before noticing the play. No injury occurred. The bolt was re-torqued and inspected by a second technician

the same afternoon, and T-22 was cleared for climbing again. This near-miss has been formally logged and is discussed further, including a site-wide policy change, in the Safety Procedure bulletin and the Q3 Operations Review minutes.

## 7. Lubrication & Consumables

Routine gearbox oil top-ups were completed on 6 turbines per the standard schedule. Hydraulic filter elements were replaced on turbines T-02, T-09, and T-14 as part of the quarterly consumables rotation. Grease reservoirs for pitch and yaw bearings were topped up site-wide; no abnormal consumption rates were observed.

## 8. Scheduled Maintenance - Next Two Weeks

| Date   | Turbine    | Task                                            | Owner                      |
|--------|------------|-------------------------------------------------|----------------------------|
| Sep 16 | T-19       | OEM gearbox inspection follow-up                | Senior Mechanical Engineer |
| Sep 18 | T-04, T-08 | Quarterly blade inspection                      | Mechanical Team B          |
| Sep 22 | T-07       | Brake pad replacement                           | Mechanical Team A          |
| Sep 25 | T-22       | Anchor point re-certification (full site audit) | Site Safety Officer        |

## 9. Action Items Summary

- CRITICAL - Senior Mechanical Engineer: obtain a procurement decision on expedited vs. standard gearbox replacement for T-19 by September 16, 2026.
- HIGH - Mechanical Team A: replace T-07 brake pads by September 22, 2026.
- HIGH - Site Safety Officer: complete a full anchor-point re-certification audit across all turbines by September 25, 2026, following the T-22 near-miss.
- MEDIUM - Electrical Team: continue monitoring T-03 transformer oil weep; reseal if weep exceeds 100ml before the next inspection.
- LOW - Operations: no action required on T-11 yaw bearing noise; re-check at next scheduled inspection.

## 10. Sign-off

Prepared by: J. Alvarez, Mechanical Maintenance Lead. Reviewed by: R. Okafor, Site Operations Manager. Distribution: Operations Management, Health & Safety, Finance (T-19 cost exposure).